



Solve with us 2.2 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1

A matrix is in row echelon form if :

- The first non-zero element (the leading entry) in a row is 1.
- The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.
- Any non-zero rows are always above rows with all zeros.

A matrix is in reduced row echelon form if :

- The first non-zero element in the first row (the leading entry) is the number 1.
- All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.
- The leading entry in each row must be the only non-zero number in its column.
- Any non-zero rows are always above rows with all zeros.

Note: Any matrix which is in reduced row echelon form is also in row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $A = \begin{bmatrix} & & \\ 1 & 0 & 2 & 0 & 0 & 1 & 0 \\ & & & & & & \end{bmatrix}$

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA is in both row echelon form and reduced row echelon form.

1 point

Choose the correct options for the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ $A = \begin{bmatrix} & & \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ & & & & & & \end{bmatrix}$

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.



No, the answer is incorrect.

Score: 0

Accepted Answers:

AA is neither in row echelon form nor reduced row echelon form.

1 point

Choose the correct options for the matrix $F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA is in row echelon form but not in reduced row echelon form.

Key Points: 2

Let $Ax = b$ be a system of linear equations and suppose AA is in reduced row echelon form. Assume that for every zero row of A , the corresponding entry of b is also 0 (i.e., if the i -th row of AA is 00, then so is b_i .)

- If the i -th column has the leading entry of some row, we call x_i a dependent variable.
- If the i -th column does not have the leading entry of any row, we call x_i an independent variable.

1 point

Consider a system of linear equations:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned} \quad \begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 10 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned}$$

Choose the set of correct options.

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

☐

x_1 and x_2 are dependent variables.

☐

x_2 is a dependent variable.

☐

x_3 and x_4 are independent variables.

☐

x_4 is an independent variable.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_2 is a dependent variable.

x_4 is an independent variable.

1 point

let $Ax = b$ be a system of linear equations, where $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, and

$$b = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

Choose the set of correct options.

☐

x_1 is a dependent variable.

☐

x_2 is a dependent variable.

☐

x_3 is a dependent variable.

☐

The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

x_1 is a dependent variable.

x_3 is a dependent variable.

The expressions $x_1 = 3$ and $x_3 = 2 - x_4$ show the dependency of x_1 and x_3 on the elements of b and the independent variables.

Key Points: 3

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$A = \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

1 point

What will be the third row of the matrix if we perform the row operation $R_3 - 2R_1$ on A ?

☐

(1123)

☐

(41083)(41083)

☐

(0-203)(0-203)

☐

(0443)(0443)

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(0-203)(0-203)

*1 point*What will be the second row of the matrix if we perform the row operation $\frac{1}{2}R_2 \rightarrow 2R_2$ on AA ?☐

(0110)(0110)

☐

(0420)(0420)

☐(01 $\frac{1}{2}$ 0)(01210)☐(122 $\frac{3}{2}$)(12223)**No, the answer is incorrect.****Score: 0****Accepted Answers:**(01 $\frac{1}{2}$ 0)(01210)*1 point*If we perform the row operations mentioned in question (a) and question (b), simultaneously on AA , the matrix (let it be called BB) we get is:☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 1 & \frac{1}{2}0 \\ 0 & -203 \end{bmatrix} \begin{bmatrix} 10031-22210003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1320 \\ 0110 \\ 1123 \end{bmatrix} \begin{bmatrix} 101311212003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 4 & 20 \\ 0 & -203 \end{bmatrix} \begin{bmatrix} 10034-2220003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1320 \\ 01\frac{1}{2}0 \\ 0443 \end{bmatrix} \begin{bmatrix} 1003142214003 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 1 & -\frac{1}{3} \\ 0 & -20 & 3 \end{bmatrix} \begin{bmatrix} 10031-22210003 \\ 10031-22210003 \\ 10031-22210003 \end{bmatrix}$$

1 point

Which row operations have to be performed on the matrix BB (matrix BB is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row.

☐

$R_1 - 3R_2$ $R_1 - 3R_2$ and $R_3 - 2R_2$ $R_3 - 2R_2$

☐

$R_1 + 3R_2$ $R_1 + 3R_2$ and $R_3 + 2R_2$ $R_3 + 2R_2$

☐

$R_1 + 3R_2$ $R_1 + 3R_2$ and $R_3 - 2R_2$ $R_3 - 2R_2$

☐

$R_1 - 3R_2$ $R_1 - 3R_2$ and $R_3 + 2R_2$ $R_3 + 2R_2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$R_1 - 3R_2$ $R_1 - 3R_2$ and $R_3 + 2R_2$ $R_3 + 2R_2$

1 point

The reduced row echelon form of AA is

[Hint: Recall the definition of reduced row echelon form.]

☐

$$\begin{bmatrix} 10 & 10 \\ 01 & -\frac{1}{3} \\ 00 & 13 \end{bmatrix} \begin{bmatrix} 10001021211003 \\ 10001021211003 \\ 10001021211003 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1000 \\ 0100 \\ 0013 \end{bmatrix} \begin{bmatrix} 100010001003 \\ 100010001003 \\ 100010001003 \end{bmatrix}$$

☐

$$\begin{bmatrix} 100 & 0 \\ 010 & -\frac{3}{2} \\ 001 & 3 \end{bmatrix} \begin{bmatrix} 1000100010-233 \\ 1000100010-233 \\ 1000100010-233 \end{bmatrix}$$

☐

$$\begin{bmatrix} 100 & -\frac{3}{2} \\ 010 & -\frac{3}{2} \\ 001 & 3 \end{bmatrix} \begin{bmatrix} 100010001-23-233 \\ 100010001-23-233 \\ 100010001-23-233 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 100 & -3 \\ 010 & -3 \\ 001 & 3 \end{bmatrix} \begin{bmatrix} 100010001 & -23 & -233 \end{bmatrix}$$

Key Points: 4

Let $Ax = b$ denote the matrix representation of a system of linear equations.

- The augmented matrix is denoted by $[A | b]$.
- The matrix obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix is denoted by $[R | c]$.
- The solutions of $Rx = c$ are the same as the solutions of $Ax = b$.

Consider a system of linear equations

$$\begin{aligned} 2x_1 + x_2 &= 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1. \\ 2x_1 + x_2 &= 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1. \end{aligned}$$

Answer the questions (a), (b) and (c) based on the above information.

1 point

Which of the following represents an augmented matrix of the system?

[Hint: First try to write down the coefficient matrix A and the corresponding vector b .]

- ☐
- ☐
- ☐
- ☐

No, the answer is incorrect.

Score: 0

Accepted Answers:

1 point

What will be the matrix $[R | c]$ obtained by performing the operations that transform A to its reduced row echelon form R on the augmented matrix?

- ☐
- ☐
- ☐
- ☐

No, the answer is incorrect.

Score: 0

Accepted Answers:

1 point

Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.]

- ☐ The system of linear equations has an infinite number of solutions.
- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐

$x_1 = x_2 = x_3 = x_4 = 1$ is a solution of the system of linear equations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has no solution.

[Check Answers](#)